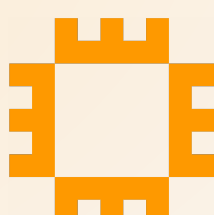


WAYANAMAC EDUCATION TRUST®

Don Bosco Institute of Technology

MYSORE ROAD, KUMBALAGODU, BANGALORE - 560074, KARNATAKA
Affiliated to VTU - Approved by AICTE



DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

AWS STUDENT BUILDER GROUP

Grid2Cloud: From Engineering Problems to Cloud Solutions

LEARN // EXPLORE // CONNECT // BUILD

1 DAY - BEGINNER-FRIENDLY - 2 & 3 OCT 2026

Welcome to AWS × EEE

CLOUD BASICS → AWS FOR EEE → CORE SERVICES → MESSAGING & MONITORING → SERVICE WORKFLOW → DATA TO ACTION → MINI PROJECT → SOLUTION DESIGN → YOUR OWN AWS SOLUTION

PREREQUISITES

Before You Arrive

- Basic understanding of Electrical and Electronics Engineering concepts
- Active internet connection
- AWS account for hands-on activities
- Basic programming knowledge is helpful but not mandatory
- No prior AWS experience required

WORKSHOP BRIEF

BEGINNER-FRIENDLY // HANDS-ON // 1 DAY

This one-day hands-on workshop introduces students to cloud computing and Amazon Web Services (AWS) through an Electrical and Electronics Engineering perspective. Participants learn how cloud services can be used to collect sensor data, process information, store data, monitor systems, and generate alerts for real-world EEE applications. The workshop covers AWS IoT Core, SQS, Lambda, CloudWatch, S3, EC2, SNS, and Amazon MQ, followed by a practical mini-project and student solution-design activity.

SESSION 1

Understanding Cloud Computing & AWS

- Introduction to cloud computing and traditional vs. cloud infrastructure
- Benefits of cloud computing
- Introduction to AWS
- How cloud technology can support EEE systems

Checkpoint: understand cloud computing and AWS fundamentals

SESSION 2

AWS for EEE Applications

- Smart energy systems, industrial automation, renewable energy
- Predictive maintenance and smart homes
- Students identify engineering problems cloud technology can solve

Checkpoint: identify at least one EEE problem suitable for a cloud-based solution

SESSION 3

AWS Core Services

- Introduction to AWS IoT Core, AWS Lambda, Amazon S3, and Amazon EC2
- What each service does and when to use it
- How services work together, with simple demonstrations

Checkpoint: understand the purpose and use cases of the four core AWS services

SESSION 4

Messaging & Monitoring Services

- Hands-on introduction to Amazon SQS, SNS, CloudWatch, and Amazon MQ
- How engineering systems handle messages
- Monitoring performance and sending alerts on abnormal conditions

Checkpoint: match messaging, monitoring, and alerting needs to the right AWS services

SESSION 5

Connecting AWS Services

- Sensor → IoT Core → SQS → Lambda → S3 → CloudWatch → SNS → Engineer/Action
- Workflow explained using an EEE-based example

Checkpoint: create and understand a complete AWS service workflow

SESSION 6

From Data to Engineering Action

- Examining sample data — voltage, current, energy consumption, temperature
- Processing, storing, and monitoring raw sensor data
- Converting data into engineering decisions and alerts

Checkpoint: interpret sample engineering data and determine appropriate actions

SESSION 7

Student Challenge & Solution Design

- Select a practical EEE problem
- Design an AWS-based solution and required data
- Draw system architecture and explain the data flow
- Prepare a short presentation of the proposed solution

Checkpoint: produce a problem statement, AWS architecture, service selection, data flow, and short solution explanation

KEY TAKEAWAYS

What You'll Walk Away With

- Understand the fundamentals of cloud computing and AWS
- Learn how AWS applies to Electrical and Electronics Engineering
- Understand AWS IoT Core, SQS, Lambda, CloudWatch, S3, EC2, SNS, and Amazon MQ
- Learn how different AWS services communicate and work together
- Understand the journey from sensor data to engineering action
- Learn basic cloud architecture and service-selection thinking
- Design an AWS-based solution for a real-world EEE problem
- Gain hands-on exposure to cloud-based engineering workflows
- Develop a complete mini-project concept using AWS services

CERTIFICATE

AWS RESOURCE MATERIAL

MINI-PROJECT REFERENCE

ARCHITECTURE & SERVICE SHEET

DESIGN YOUR OWN AWS-POWERED EEE SOLUTION

WORKSHOP GUIDE

AWS × EEE — From Engineering Problems to Cloud Solutions // Department of Electrical and Electronics Engineering (EEE) // Date To Be Announced

SESSION	TOPIC	OUTCOME
01	Understanding Cloud Computing & AWS	Understand cloud computing and AWS fundamentals
02	AWS for EEE Applications	Identify an EEE problem suitable for a cloud solution
03	AWS Core Services	Understand IoT Core, Lambda, S3 & EC2 use cases
04	Messaging & Monitoring Services	Match messaging & alerting needs to AWS services
05	Connecting AWS Services	Build a complete AWS service workflow
06	From Data to Engineering Action	Interpret engineering data into actions
07	Student Challenge & Solution Design	Produce an original AWS-based EEE solution design

AWS × EEE — From Engineering Problems to Cloud Solutions

A one-day, beginner-friendly workshop that introduces cloud computing and AWS through an Electrical and Electronics Engineering lens — from sensor data and messaging to monitoring and alerts. Participants work through AWS IoT Core, Lambda, S3, EC2, SQS, SNS, CloudWatch, and Amazon MQ, then apply what they've learned to a real Smart Energy Monitoring mini-project and their own solution design.

QUICK OVERVIEW — THE FLOW

1 Day
DURATION

7
SESSIONS

Beginner-Friendly
SKILL LEVEL

Certificate Issued
OUTCOME

Key Takeaways

Fundamentals of cloud computing and AWS

How AWS applies to Electrical and Electronics Engineering

AWS IoT Core, SQS, Lambda, CloudWatch, S3, EC2, SNS & Amazon MQ

How different AWS services work together

The journey from sensor data to engineering action

Basic cloud architecture and service-selection thinking

Designing an AWS-based solution for a real EEE problem

Hands-on exposure to cloud-based engineering workflows

Coordinators

STUDENT COORD.

Varsha B

Dept. of EEE

Poorvika BN

Dept. of EEE

Mayur R

Dept. of EEE

Moksha NS

Dept. of EEE

FACULTY COORD.

Dr. Jyothi Roogi

Associate Professor, EEE

ORGANIZING TEAM

Lohith G Gowda

AWS SBG Core Member

Priyadarshini K

AWS SBG Core Member

For queries, contact the AWS Student Builder Group Team @ Lohith G Gowda: 8884056006 / Priyadarshini K: 8792842818

@AWSSBG_DBIT // AWS STUDENT BUILDER GROUP